we are trying to work on a project whose detail are given below, for the module 4. Detecting Greenwashing in Sustainability Reports Using NLP we want to identify 5 interesting research questions that fall in the same domain as this topic and a typical IEEE reference. Detecting Greenwashing in Sustainability Reports Using NLP Title: AI-Based Greenwashing Detection in Corporate Sustainability Reports Fundamental idea: Students use NLP techniques to analyse corporate sustainability text and identify vague, exaggerated, or low-evidence environmental claims. Example 1: Collect sustainability report paragraphs and classify them into concrete claims, vague claims, and unverifiable claims.

展开

AI-Based Greenwashing Detection in Corporate Sustainability Reports

Introduction

Greenwashing refers to the practice of presenting an environmentally responsible image without sufficient evidence to support sustainability claims. As Environmental, Social, and Governance (ESG) reporting becomes increasingly important, companies publish sustainability reports to communicate their environmental performance. However, some disclosures contain vague, exaggerated, or unverifiable statements that may mislead stakeholders.

This project applies Natural Language Processing (NLP) techniques to analyse sustainability reports and detect potential greenwashing practices. By examining textual patterns, disclosure quality, commitment-action consistency, and linguistic features, the study aims to provide a data-driven framework for

This project applies Natural Language Processing (NLP) techniques to analyse sustainability reports and detect potential greenwashing practices. By examining textual patterns, disclosure quality, commitment-action consistency, and linguistic features, the study aims to provide a data-driven framework for evaluating the credibility of corporate sustainability communication.

Research Question 1 (RQ1)

What is the distribution of concrete, vague, and unverifiable environmental claims in corporate sustainability reports?

Motivation

Environmental disclosures vary in quality and credibility. Some claims provide measurable evidence, while others rely on general statements without supporting information. Understanding the distribution of different claim types helps assess the prevalence of potential greenwashing.

Methodology

- Text preprocessing and cleaning
- Claim extraction from sustainability reports
- NLP-based classification into:
  - Concrete Claims
  - Vague Claims
  - Unverifiable Claims

Expected Outcome

The analysis will reveal how frequently each type of claim appears and identify whether vague or unverifiable statements dominate corporate sustainability communication.

Research Question 4 (RQ4)

How do green-related keywords and linguistic patterns differ between concrete, vague, and unverifiable sustainability claims?

Motivation

Language plays a significant role in greenwashing. Companies may use environmentally positive terminology without providing supporting evidence.

Methodology

- Keyword frequency analysis
- Text mining
- Comparative linguistic analysis
- Green-related vocabulary extraction

Expected Outcome

The study will identify linguistic patterns associated with different claim categories and reveal how environmental terminology is used across disclosure types.

Research Question 5 (RQ5)

Which linguistic features are the most important predictors of potential greenwashing in sustainability reports?

Motivation

Understanding the factors that influence greenwashing detection improves the transparency and interpretability of NLP models.

Methodology

Research Question 2 (RQ2)

Can NLP techniques detect inconsistencies between companies' environmental commitments and their reported sustainability actions?

Motivation

Many companies announce ambitious climate goals such as carbon neutrality or net-zero emissions. However, the actions reported in sustainability reports may not provide sufficient evidence of progress toward these commitments.

Methodology

- Sentence embeddings using BERT or Sentence-BERT
- Semantic similarity analysis
- Commitment-action matching
- Inconsistency scoring

Expected Outcome

The study will identify gaps between stated environmental commitments and reported actions, highlighting potential indicators of greenwashing.

Research Question 3 (RQ3)

Do companies with higher ESG disclosure intensity provide more specific and evidence-based environmental disclosures?

Motivation

A larger volume of ESG disclosures does not necessarily indicate higher transparency or accountability. This question investigates whether extensive reporting is associated with greater disclosure quality.

Research Question 4 (RQ4)

How do green-related keywords and linguistic patterns differ between concrete, vague, and unverifiable sustainability claims?

Motivation

Language plays a significant role in greenwashing. Companies may use environmentally positive terminology without providing supporting evidence.

Methodology

- Keyword frequency analysis
- Text mining
- Comparative linguistic analysis
- Green-related vocabulary extraction

Expected Outcome

The study will identify linguistic patterns associated with different claim categories and reveal how environmental terminology is used across disclosure types.

Research Question 5 (RQ5)

Which linguistic features are the most important predictors of potential greenwashing in sustainability reports?

Motivation

Understanding the factors that influence greenwashing detection improves the transparency and interpretability of NLP models.

Methodology

With these research questions and the idea in place, can you identify publicly available dataset suitable for these research questions?

Upload the selected and downloaded datasets. Then give this prompt.

We select Public Company ESG Ratings Dataset and S&P 500 ESG Sustainability Reports Dataset. For each research question, please identify a publication ready figure and a publication ready table for depicting the answer for the research question. Generate the

展开

Since your project already uses:

1. **Public Company ESG Ratings Dataset**
2. **S&P 500 ESG Sustainability Reports Dataset**

these two datasets are sufficient to answer all five research questions.

Dataset Selection

Dataset 1: Public Company ESG Ratings Dataset

Purpose

Used for:

- RQ2 Commitment vs Action Inconsistency
- RQ3 ESG Disclosure Intensity vs Specificity

Main Variables

Variable	Description
Company	C any name

Dataset 2: S&P 500 ESG Sustainability Reports Dataset

Purpose

Used for:

- RQ1 Claim Classification
- RQ2 Commitment vs Action Analysis
- RQ4 Green Keyword Analysis
- RQ5 NLP Feature Analysis

Main Variables

Variable	Description
Company	Company name
Report Year	Reporting year
Paragraph	Sustainability disclosure text
Climate Commitment	Target statements
Reported Action	Action statements
Claim Type	Concrete / Vague / Unverifiable
NLP Features	Linguistic characteristics

RQ1

Research Question

What is the distribution of concrete, vague, and unverifiable environmental claims in corporate sustainability reports?

Suppose we want to run the code on Kaggle. Now for each research question, I want a separate ipynb file. It should save all figures as pdfs and all tables as csvs. Now generate the five ipynb files (to be uploaded and ran from Kaggle) that take the raw data as input and give us the actual resulting tables and figures from the input, by implementing the methodology defined earlier.

收起

你这个要求实际上是在让生成 **5个完整的 Kaggle Notebook（.ipynb）项目代码**。但目前我还缺少一个关键条件：

我需要知道你最终使用的数据文件名称和字段名

根据我们之前的聊天，我知道你有：

- `emerald_data.csv`
- `preprocessed_content.csv`
- Public Company ESG Ratings Dataset
- S&P 500 ESG Sustainability Reports Dataset

但是我不知道 Kaggle 上最终上传的是：

```
/input/
    esg_ratings.csv
    sustainability_reports.csv
```

还是：

```
/input/
    emerald_data.csv
    preprocessed_content.csv
```

以及字段名是否是：

Python 运行

```python
company
text
claim_type
```

有问题，尽管问

I want to generate a scientific publication ready step by step workflow figure for this topic that is illustrative of the whole process, uses colors and numbers each step, in the same way as the attached workflow figure.

有问题，尽管问

ChatGPT 也可能会犯错。请核查重要信息。